When Is a Reconciliation Break Real? A Temporal Correctness Contract for Streaming Financial Reconciliation

Abhishek Sharma
Senior Member, IEEE
abhicse24@gmail.com ORCID: 0009-0007-1103-2103

***Abstract*—Reconciliation systems decide, continuously, whether money the ledger expected has arrived at the bank. On an asynchronous stream that decision is taken against partial evidence: a credit that has not appeared may be missing, or may simply not have been posted yet. Deployed pipelines settle this with a timeout or a watermark, and the resulting exceptions are treated as findings. We give a temporal correctness contract that says when such a verdict is admissible, relative to a declared per-source lateness bound, and a checker for it that needs no ground truth. The contract separates three things an operator otherwise sees only as a rising break rate: the reconciler decided too early, an input broke the bound it declared, or the money really is gone. On workloads built from a published payment simulator we find that the standard remedy for a stalled streaming join, an idleness timeout, introduces contract violations on 5.42% of obligations (95% CI [3.94, 6.92]) on traffic where the same job without it is exactly sound, and that violations vanish once the timeout reaches the slow source's posting period, as predicted. Under a lateness shift, all four identity-based reconcilers we test raise spurious breaks on between 18.3% and 23.0% of the credits the bank did settle, while their contract violation rate stays at or below 1.01%: the checker attributes the loss to 240 bank-side and 413 ledger-side bound violations rather than to the reconcilers. Tracking one horizon per source, rather than one over the merged stream, leaves break latency 12.7% lower on average and halves its tail (p99 224 against 446 ticks) at equal soundness, though it is not cheaper at every cost setting and we report where it loses. An Apache Flink interval join, written the ordinary way, violates the contract on 14.9–20.2% of obligations, reproducibly across two independent runs of the whole experiment. The contract is expressible as a SQL query over the tables a reconciliation system already keeps, and the query and the reference implementation agree on all 4,395 verdicts compared.**

***Index Terms*—stream processing, financial reconciliation, temporal semantics, out-of-order events, watermarks, data quality**

I. Introduction

A payment settles, and two systems find out at different times. The ledger learns within seconds, because the processor told it. The bank feed posts on a cycle, so the same fact can take an hour to appear. In between, a reconciliation system looking for the credit sees nothing, and has to decide what nothing means.

Most of the time it means the batch has not run. Occasionally it means the money is gone. The two are indistinguishable from the evidence available at the moment the question is asked, and the distinction is the whole job.

What deployed systems do is pick a number. Wait ninety minutes, then raise an exception; or set a watermark and let the stream processor decide when the window has closed. Both are defensible engineering and both are, as correctness criteria, unstated: the number encodes an assumption about how late the bank can be, that assumption is nowhere written down, and when the break rate doubles there is no way to tell whether the reconciler became wrong or the bank became slow. The exception queue grows either way, and the team retunes the timeout either way.

This paper writes the assumption down and makes it checkable. The contribution is a temporal correctness contract for reconciliation on asynchronous streams: a statement of when a reconciler is *entitled* to declare a match or a break, given only what it has seen and a declared bound on how late each source may be. Entitlement is not accuracy. A reconciler can be entitled and wrong, which happens when an input violates the bound it declared, and it can be unentitled and right, which happens whenever the traffic was benign. Those two cases have different owners and different fixes, and telling them apart does not require knowing the answer, which is what makes the check deployable.

Three findings follow from the contract, and one of them is a negative result about our own construction.

The documented fix for a stalled streaming join is unsound for reconciliation. A merged watermark takes the minimum across inputs, so a source that posts in batches freezes the operator between batches; the remedy is an idleness timeout that drops the quiet input from the minimum. For a reconciliation join the quiet input is the one the decision is about. We show the resulting violations appear exactly when the idleness timeout falls below the posting period and not above it, on traffic where the same job is otherwise exactly sound.

Under a shift in the lateness distribution, entitlement and accuracy come apart cleanly. Every arm we test raises spurious breaks on between 18.3% and 23.0% of the credits that did settle, while its contract violation rate stays at or below 1.01%. The checker attributes the degradation to the inputs, and it does so from the stream and the declared model alone.

Tracking a horizon per source is better on latency and not uniformly better on cost. At matched soundness it halves the tail of break latency and reduces its standard deviation from 60.8 to 5.0 ticks, because a timeout measured from the ledger's

report inherits the ledger's own jitter into a decision that is about the bank. At the calibrated operating point, on the other hand, it decides earlier and is therefore more exposed when bounds are violated: over a $5\times5\times5$ sweep of the cost constants it is cheapest on all 125 points in calm traffic and on only 30 of 125 under faults. We report both.

II. Background and related work

A. Monitoring temporal properties over out-of-order streams

Basin, Klaedtke and Zălinescu give a runtime verification framework for metric temporal logic with freeze quantifiers over streams whose events arrive out of order, using a three-valued semantics in which a verdict may be not yet determined [1], [2]. Their algorithms are sound and complete, and POLÍMON [3] continues that line with a tool that reports verdicts promptly. The machinery this paper needs is theirs: an undetermined verdict under partial knowledge is exactly their third value.

What a general framework leaves open is the formula. For reconciliation the property turns on something neither a temporal logic nor a stream processor represents: which *source* an absence of evidence is about. The safety property below is not "a match eventually occurs". It is that silence from the bank carries no information until the bank's own completeness horizon has passed, and that horizon is a property of the bank rather than of the stream the reconciler is reading.

B. Reconciliation benchmarks

FinBalance [4] assembles multi-document accounting bundles with labelled inconsistencies; FinRCA-Bench [5] builds accounts-payable-to-bank cases and evaluates evidence retrieval against typed provenance graphs. Both present a model with a complete evidence set and ask for the right answer, and both are well suited to that question. Neither has a time axis along which evidence is still arriving, so neither can express a verdict that was correct but premature, which is the object of study here.

C. Temporal stream models and speculation

The CEDR line of work separates valid time, occurrence time and system time and defines consistency levels over speculative output with retraction [6]. Retraction is property M1 below and is orthogonal to the safety properties: a system that may retract still needs to know whether its first verdict was justified, and in reconciliation retraction is often unavailable because a notice has already gone out.

D. Industrial practice

Reconciliation over unbounded streams has been patented in the form of injected window and shard sentinels that delimit an unbounded stream into reconcilable units, with producer-defined rules carried in the closing sentinel [7]. That is an implementation technique for bounding work, complementary to a correctness property; it states no temporal condition on when a verdict is admissible.

Practitioner writing on exception management describes the phenomenon accurately, noting that an item inside its settlement window is pending and not a break, and recommends holding such items until the window expires. That instinct is the safety property below in informal clothes. What is missing is a statement of the property, a way to check a pipeline against it, and a measurement.

E. Payment agents and effect-level evaluation

FinalityBench [8] evaluates agents that act on payment outcomes before those outcomes are determined, and supplies the simulator, fault engine and delivery streams used here. Its concern and this one are the same mistake at opposite ends of the pipeline: an agent shipping goods against a capture that has not settled, and a reconciler raising a break against a credit that has not posted. CausalLoss-Fin [9] attributes realised loss to infrastructure or policy after the fact using counterfactuals; that analysis needs ground truth, where the decomposition here does not. Operation identity, what makes two records the same money movement, follows the provider-aware contract of [10].

III. The contract

A. Setting

Two evidence sources publish records derived from the same underlying money movements. The *ledger* reports what the business believes it is owed; the *bank* reports what actually moved. Write $ev(r)$ for the instant the movement described by record r occurred and $at(r)$ for the instant its source published it. The two are never collapsed: every property below relates them.

Definition 1 (Source model)**.** *A source model M assigns to each source s a lateness bound Λ_s, meaning $at(r) - ev(r) \le \Lambda_s$ for every record r from s, and a settlement window W during which a counterpart movement may legitimately occur. The* horizon *of s at time t is $H_s(t) = t - \Lambda_s$: every event of s with event time at or below $H_s(t)$ has arrived by t, if it was ever going to.*

A source model is a declaration, not a fact. An input that exceeds its bound has broken the interface, and the contract is relative to the declaration. We return to that, because it is what makes the check useful instead of merely strict.

An *obligation* o is a money movement the ledger has reported, with identity $(\text{case}, \text{reference}, \text{kind})$ following [10] and event time $ev(o)$. At each instant a reconciler assigns every obligation one of PENDING, MATCHED or EXCEPTION. Write

$$D(o) = ev(o) + W + \Lambda_{\text{bank}} \tag{1}$$

for the first instant at which the bank's horizon has passed the last moment a counterpart could have occurred.

B. Four properties

a) S1, no premature break: A reconciler may assign EXCEPTION to o at time t only if $t \ge D(o)$. Before that instant, silence from the bank is consistent with a continuation in which the credit posts, so an exception asserts something the evidence does not support.

b) S2, no premature match: A reconciler may assign MATCHED to o at t only if no admissible continuation makes that pairing wrong. A pairing justified by identity satisfies this on arrival: nothing that arrives later makes a record describe a different movement. A pairing justified by attributes, an amount and a nearby date, does not, because a later record may turn out to be the real counterpart, and is admissible only once $t \geq D(o)$.

c) L1, decisiveness: Every obligation receives a terminal verdict within a bounded delay of $D(o)$. Without it, never deciding would satisfy both safety properties.

d) M1, stability: A terminal verdict is not retracted.

C. A worked instance

Take the simulator's own numbers. A capture settles at tick 1000. The ledger publishes at 1002, so an obligation exists from then. The bank posts on a sixty-tick cycle, no sooner than fifteen ticks after the money moved, so without knowing where in that cycle the movement fell the credit can appear anywhere in $[1015, 1074]$: as prompt as the lag, or one tick short of a further full cycle. That range is what a reconciler may assume, and it gives $\Lambda_{\text{bank}} = 75$. With $W = 0$, (1) gives $D(o) = 1075$.

This movement in fact posts at 1020, twenty ticks after settling, and the distinction matters more than it looks. The contract is stated over what a reconciler can know, and where in the batch cycle a particular movement landed is not among it. A reconciler that happened to be right about 1020 would have been guessing.

A reconciler that raises a break at 1060 is wrong to do so even if the credit never comes: at 1060 the declared bound had not expired, and a continuation exists in which the credit posts at 1070. The verdict may turn out correct, and it was not justified. That is the distinction the contract makes, and the reason it can be checked without knowing the answer.

Now suppose the ledger itself was late and published at 1200. A timeout of 80 ticks measured from publication fires at 1280, which is 205 ticks after the decision became sound. The same timeout on a promptly reported movement fires at 1082. Same bound, same bank, decisions 198 ticks apart, and the difference is entirely attributable to a source that has nothing to do with the question being asked.

D. Two consequences

The first proposition explains a variance result rather than a mean one, and is the reason a per-source horizon is worth the complexity.

Proposition 1 (Ledger-lag inheritance)**.** *Let a reconciler raise an exception for o at $raised(o) + T$, where $raised(o)$ is the arrival instant of the ledger record and T is a fixed timeout. Its closure delay $raised(o)+T-D(o)$ equals $\lambda(o)+T-W-\Lambda_{\text{bank}}$, where $\lambda(o) = at(o) - ev(o)$ is the realised ledger lag. The closure delay therefore has the variance of the ledger lag distribution, for every T. A reconciler that decides at $ev(o)+B$ for a declared bound B has closure delay $B - W - \Lambda_{\text{bank}}$, a constant.*

Proof. Immediate from $raised(o) = ev(o)+\lambda(o)$ and (1). The second statement follows because $ev(o)$ is carried in the record and does not depend on when it arrived. □

Raising the timeout shifts the whole distribution and does not narrow it. An operator who wants a predictable exception queue cannot get one by tuning T.

Proposition 2 (Idleness and soundness)**.** *Let the bank publish in batches of period P, so it emits nothing for up to P between batches, and let a reconciler decide from a merged watermark $\min_s(\max\{ev(r) : r \text{ from } s \text{ seen}\}) - \delta$ with an idleness timeout τ that removes from the minimum any source silent for longer than τ. If $\tau < P$ there are inter-batch gaps during which the bank is marked idle although it is behaving normally; the watermark is then determined by the remaining sources and can exceed $H_{\text{bank}}(t)$, so S1 can be violated. If $\tau \geq P$ the bank is never marked idle merely for being between batches.*

Proof. A gap of length g with $\tau < g \leq P$ occurs whenever consecutive batches are that far apart. During such a gap the bank contributes nothing to the minimum, which is then $\max_s$ over the remaining sources less δ. The ledger publishes continuously and with smaller lateness, so its maximum event time advances during the gap and the merged watermark advances with it, without any bound tying it to Λ_{bank}. Any obligation whose event time falls below the advanced watermark is assigned an exception at a t that need not satisfy $t \geq D(o)$. □

Proposition 2 makes a specific, falsifiable prediction: violations appear below $\tau = P$ and not above it. Section VII-E tests it, and Section VII-I tests it on Apache Flink rather than on our model of one.

IV. Checking the contract

A. Two axes, kept apart

The checker scores each terminal verdict twice. *Admissibility* asks whether the verdict satisfied S1 and S2 at the instant it was emitted, and is computed from the verdict, the obligation and the source model, with no ground truth. *Outcome* asks whether it was right, and needs the oracle.

Neither implies the other, and the gap is the point. A reconciler that raises a break the moment a credit is overdue is inadmissible, and on a stream where nothing was ever late it is also correct every time; its accuracy is a property of the traffic. A reconciler can equally be admissible and wrong, which happens when a source exceeds its declared bound. The checker reports those separately and does not charge them to the reconciler, because a contract relative to a source model that blamed the reconciler for a broken model would be unfalsifiable.

B. The contract as a query

Reconciliation data is already in a database, so the contract is stated twice: once as the predicates above, and once as SQL over three tables: arrivals, obligations and verdicts.

Admissibility is a join and a comparison against (1); the source-model check is an aggregate over arrivals. No window function or recursive query is needed, which is part of the argument that the check is cheap enough to run continuously.

M1 is deliberately not in the query. It constrains a history rather than a row, and a per-verdict join would misstate it; retractions are counted separately.

V. Reconcilers

Five arms, four of them drawn from practice.

Fixed timeout raises a break when no counterpart arrives within T of the obligation being raised. It is what an operations team writes first, and by Proposition 1 its closure delay carries the ledger's jitter.

Merged watermark is the textbook interval join: a per-source watermark of $\max ev - \delta$, and the operator takes the minimum. Sound in ordinary configurations, and not timely, because the bank's watermark only moves at a batch boundary.

Merged watermark with idleness is that job after someone fixes the stall, with the idleness timeout of Proposition 2.

Amount/date matching pairs on equal amount and proximity in time, modelling deployments with no shared reference. Its failure is ambiguity rather than lateness.

Typed horizon is the contract's reference implementation. It declares a bound per source and matches on identity, so a match is asserted on arrival, and it waits for the *bank's* horizon specifically before raising a break. Duplicate records are dropped by identity rather than matched again.

VI. Experimental setup

Workloads are built from FinalityBench [8], which supplies a canonical event log per case, a fault engine over deliveries, and per-system delivery streams. Its ledger trails a settlement by two ticks; its bank posts on a 60-tick cycle no sooner than 15 ticks after the money moves, so a fault-free fact can wait 75 ticks. That heterogeneity is the research problem and is taken from the simulator rather than imposed.

Each regime uses five seeds of 200 interleaved cases spread over a 1440-tick day: 1,733 arrivals and 885 obligations in calm traffic, 1,684 and 593 under faults. Genuine breaks, movements whose bank evidence is removed outright, are injected in 45 cases, drawn before any fault is applied, because without a positive class a reconciler that never raises a break would score perfectly.

Injection fixes the class but does not hold it constant, and the difference matters for how rates are reported. In calm traffic those 45 cases yield 37 graded breaks; under faults there are 210, because delivery loss destroys further bank evidence and a credit that was dropped in transit is a real break. Spurious breaks are therefore reported throughout as a share of the obligations the bank *did* settle, 848 of them in calm traffic and 383 under faults, instead of all obligations, so that the figure does not move merely because the genuine-break class grew.

Every arm is calibrated on calm traffic by one rule: the smallest parameter with no contract violations and no spurious breaks. Nobody retunes against the regime they are scored on. Confidence intervals are percentile bootstrap over *cases*, not obligations, because obligations from one case share its amount, archetype and fault draw; comparisons use a shared case resample.

TABLE I
Reporting lag by source, in ticks, pooled over five seeds. The gap between the two sources is the reason a single horizon cannot serve both.

	source	min	median	p99	max
calm	ledger	2	2.0	2.0	2
	bank	15	40.0	73.6	74
shifted	ledger	2	2.8	234.0	265
	bank	15	58.0	283.4	326

Table I gives the lag structure. In calm traffic the ledger is exactly two ticks behind and the bank's worst case is 74, inside the declared bound of 75; the calibration procedure selects 80 for the arms that take a bound directly. Under faults both tails run far past the declaration, the ledger to 265 and the bank to 326. That is the shift the later experiments measure against.

The event store is PostgreSQL 16.4 and the stream processor is Apache Flink 1.20, both in Docker and both pinned.

VII. Results

A. What supports what

Three kinds of evidence appear below and they carry different weight, so it is worth saying once which conclusion rests on which.

Propositions are proved from the definitions and hold for any system matching them. Proposition 1 and Proposition 2 are of this kind, and no experiment can confirm or refute them; the experiments show that our implementations behave as the propositions say, which is a check on the implementations.

Simulator results are everything graded over reconcilers written for this paper on workloads from the published generator. Precision, closure delay, the frontier and the fault breakdown are all of this kind. They compare constructions against each other on identical streams, which is a like-for-like measurement, and they say nothing about any deployed system.

The engine experiment is Apache Flink 1.20 running the SQL interval join, with its own verdicts graded by the same checker (Section VII-I). It is the only arm in which a real stream processor makes a decision.

One boundary between the second and third matters enough to state before the numbers rather than after. Our watermark arms agree with Flink on roughly four obligations in five, and Section VII-I gives the reason and the figures. The timing results in the simulator are therefore properties of our models. They are not measurements of Flink, and the Flink arm is not a validation of them.

A second boundary concerns Proposition 2. It is proved, and the simulator reproduces its prediction. The engine arm does *not* test it: Flink's idleness timeout is measured in processing time and a bounded file source is consumed in milliseconds,

TABLE II

EACH ARM CALIBRATED ON CALM TRAFFIC, THEN RUN UNCHANGED ON BOTH REGIMES. "F.BREAK%" IS SPURIOUS BREAKS AS A SHARE OF THE CREDITS THE BANK SETTLED; "EXC." IS THE NUMBER OF EXCEPTIONS THE CLOSURE STATISTICS ARE COMPUTED OVER. AMOUNT/DATE MATCHING NEVER CALIBRATES AND IS SHOWN AT THE LARGEST PARAMETER TRIED.

	viol.%	f.break%	f.match%	exc.	mean	p99
calm — 885 obligations, 848 settled, no bound violations						
typed horizon	0.00	0.00	0.00	37	2.1	4
fixed timeout	0.00	0.00	0.00	37	3.8	6
merged watermark	0.00	0.00	0.00	37	32.7	223
+ idleness	0.00	0.00	0.00	37	29.9	201
amount/date	94.46	2.12	22.82	37	421.5	426
shifted — 593 obligations, 383 settled; 240 bank and 413 ledger bound violations						
typed horizon	0.17	22.98	0.00	298	12.9	146
fixed timeout	0.00	20.89	0.00	290	28.9	226
merged watermark	0.51	18.28	0.00	280	53.0	213
+ idleness	1.01	18.80	0.00	282	45.5	160
amount/date	47.05	8.36	32.72	141	458.5	637

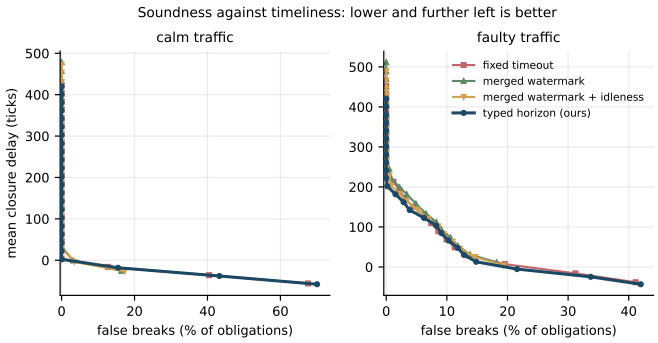


Fig. 1. Sweeping each family's parameter. Down and to the left is better.

TABLE III

CHEAPEST POINT ON EACH CURVE WITH NO SPURIOUS BREAKS, SHIFTED TRAFFIC. BRACKETS ARE 95% BOOTSTRAP INTERVALS OVER CASES.

	mean closure	p99	s.d.
typed horizon	221.6 [220.8, 222.1]	224	5.0
+ idleness	245.8 [240.6, 251.4]	392	40.4
fixed timeout	253.8 [245.7, 262.5]	446	60.8
merged watermark	264.9 [255.8, 274.2]	554	69.8

so the setting never engages, which is why its four values give identical rows in Table VI. Nothing in the engine arm confirms the proposition and nothing in it contradicts the proposition.

Intervals throughout are percentile bootstraps resampled over *cases* rather than obligations. Several obligations come from one sampled case and share its amount, archetype and fault draw, so resampling obligations would treat correlated evidence as independent and give intervals that are too narrow. Comparisons between arms use the same case resample for both.

B. Calibrated operating points

Table II gives the operating points. On calm traffic the four identity-based arms are all exactly sound, and they differ by more than an order of magnitude in how quickly a genuine break surfaces: 2.1 ticks for the typed horizon against 32.7 for the merged watermark, and a tail of 4 ticks against 223. Only 37 exceptions arise in calm traffic, every one of them a genuine break, so the calm percentile is close to a maximum and should be read as one; the shifted regime, where the statistics rest on roughly 290 exceptions per arm, is the one that supports a percentile claim. The watermark arms are not slow because they are cautious. They are slow because the bank's watermark only advances at a batch boundary, so a break discovered at the start of a cycle waits for the end of it.

Amount/date matching never calibrates. There is no parameter at which it is both sound and accurate, because its failure is not a matter of waiting: with eight distinct retail amounts in circulation, two movements of the same size in flight together are indistinguishable, and it pairs the wrong records on 22.82% of obligations in calm traffic. This is the sense in which matching rules are downstream of the temporal contract. No amount of tuning the decision deadline repairs a pairing that was never justified.

C. Soundness against timeliness

Comparing arms at their calibrated parameters is what an operator would deploy but is not a controlled comparison. Figure 1 sweeps each family; Table III reads off the cheapest point at which no spurious break is raised, which is the operating point reconciliation teams actually ask for.

The typed horizon reaches that point 12.7% sooner than a fixed timeout, with non-overlapping intervals, and its tail is halved. Figure 2 shows why that matters operationally: for the typed horizon the mean and the 99th percentile are almost the same number, so the exception queue drains predictably, while for the others the worst decile arrives at roughly twice the average delay. The standard deviation is the clearest number: 5.0 ticks against 60.8 and 69.8. That is Proposition 1 showing up in the data. The other arms inherit the jitter of the ledger's reporting into a decision that is about the bank, and raising their parameter moves the distribution without narrowing it.

D. What happens when the distribution moves

Figure 3 sweeps fault intensity from zero to 0.5 on seeds disjoint from the calibration set. The share of arrivals breaking their declared bound rises from 0% to 57.9%, and spurious breaks rise with it, reaching 25.6% of settled credits for the fixed timeout and 30.8% for the typed horizon.

The contract violation rate barely moves. Across the whole sweep it stays at or below 1.11% for every arm, and at or below 0.43% for every arm that does not use an idleness timeout. The exception is the one construction we already know breaks S1. These reconcilers were, with that caveat, entitled to the verdicts they issued, and they were wrong anyway, because the inputs stopped honouring the bounds against which they were calibrated. The checker reaches that conclusion from the

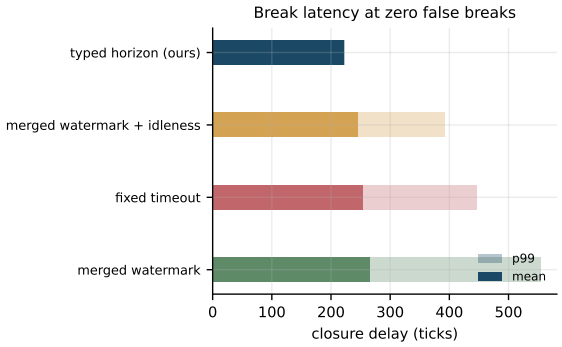


Fig. 2. Mean and 99th-percentile break latency at the zero-false-break operating point, shifted traffic. The bars for the typed horizon nearly coincide; for the others they do not.

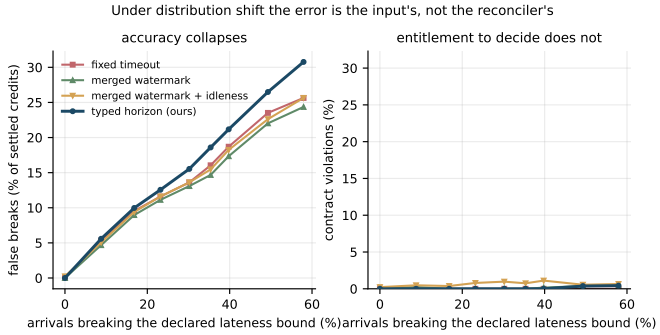


Fig. 3. As fault intensity rises, accuracy collapses and entitlement does not.

arrivals and the declared model, without consulting the oracle, which is what makes it usable on the morning the break rate triples.

This is the practical content of separating the two axes. An operator watching only the break rate sees a reconciler that has become unreliable and retunes it. The retuning cannot help: no timeout repairs a bank that has started posting late, and lengthening it trades a real detection delay for an apparent improvement.

E. The idleness trap

With idleness disabled the merged watermark commits no contract violations on calm traffic at all. Turning it on, at a timeout of 5 ticks, produces violations on 5.42% of obligations (95% CI [3.94, 6.92]) and spurious breaks on 3.05% — on traffic in which no source is ever late and every credit arrives.

Figure 4 shows the cliff. Violations fall monotonically as the timeout rises — 0.45% at 30 ticks, 0.11% at 50 — and reach exactly zero at 60, the bank's posting period, remaining there above it. That is the boundary Proposition 2 predicts, and it was predicted before it was measured.

The trap is that the correct setting defeats the purpose. Idleness is reached for because the join stalls between batches; the smallest timeout that keeps the join sound is the batch period, which is precisely the gap the operator was trying to stop waiting through.

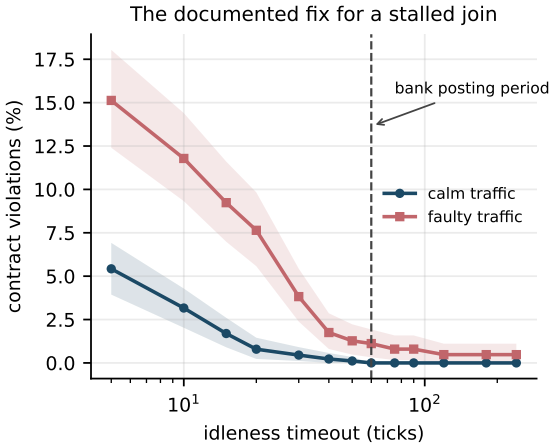


Fig. 4. Contract violations against the idleness timeout, with the bank's posting period marked.

TABLE IV
ONE FAULT FAMILY AT A TIME, AT INTENSITY 0.30. "BOUND VIOL." COUNTS ARRIVALS EXCEEDING THEIR DECLARED LATENESS. ONLY TWO FAMILIES REACH THE LEDGER-TO-BANK PATH AT ALL.

		typed horizon		amount/date
family	bound viol.	f.break%	f.match%	f.match%
none	0	0.0	0.0	24.0
delay	462	22.9	0.0	35.9
duplicate	430	0.0	0.0	36.4
loss	0	0.0	0.0	28.9
reorder	0	0.0	0.0	24.0
partial commit	0	0.0	0.0	24.0
stale read	0	0.0	0.0	24.0

F. Which faults matter

Enabling one fault family at a time is unusually decisive here. Of the six delivery-fault families in the simulator, only two touch reconciliation correctness at all. Delay produces 462 bound violations and drives spurious breaks to 22.9% for the typed horizon. Duplication produces 430 bound violations and, for identity-based arms, no errors whatsoever, because a repeated delivery of the same movement is recognised and dropped. Amount/date matching mispairs 36.4% of its verdicts under the same faults. Reordering, partial commits and stale reads leave the ledger-to-bank path untouched, with results identical to the fault-free arm to three digits.

Table IV gives the breakdown. Reconciliation correctness has a narrow failure surface: lateness and repetition. Lateness is a safety problem for everyone; repetition is a safety problem only for reconcilers that cannot tell two records apart.

G. Cost, and where our construction loses

The monetary figures rest on three constants argued rather than measured, so each is swept over an order of magnitude, giving 125 combinations per regime. On calm traffic the typed horizon is cheapest at all 125. Under shifted traffic it is cheapest at 30, and the merged watermark is cheapest at 75.

TABLE V
CHEAPEST ARM OVER A $5 \times 5 \times 5$ SWEEP OF THE THREE COST CONSTANTS, 125 COMBINATIONS PER REGIME. THE REVERSAL BETWEEN REGIMES IS THE RESULT, NOT THE DOLLAR FIGURES.

cheapest at	calm	shifted
typed horizon	125	30
merged watermark	0	75
fixed timeout	0	20
+ idleness	0	0
amount/date	0	0

TABLE VI
APACHE FLINK 1.20, SQL INTERVAL JOIN, SHIFTED TRAFFIC. EACH ROW AVERAGES EIGHT RUNS, FOUR IDLENESS SETTINGS BY TWO SEEDS; THE IDLENESS TIMEOUT CHANGES NOTHING HERE, FOR THE REASON GIVEN IN THE TEXT, SO THE SPREAD IS THE ENGINE DISAGREEING WITH ITSELF.

window	viol.%	(range)	f.break%	mean	p99
80	14.91	14.78–15.04	18.28	51.5	239
160	17.96	15.93–20.87	15.45	29.4	176
320	20.20	17.39–23.01	14.91	17.5	161

That reversal, in Table V, is real and worth stating plainly. At its calibrated parameter the typed horizon decides at the earliest instant its declared bound permits, which is exactly what makes it fast, and exactly what exposes it when the bound turns out to be wrong. The merged watermark's extra conservatism, which costs it an order of magnitude in latency on calm traffic, is what pays under shift. At matched soundness (Table III) the ordering reverses again.

The contract does not make a reconciler cheap. It makes the trade explicit, and it makes the residual attributable.

H. Checking the contract in SQL

Across five arms, three seeds and both regimes, the SQL formulation of Section IV-B and the reference implementation agree on all 4,395 verdicts compared, on both admissibility and outcome, with no disagreements. The check is a join and a comparison, so a reconciliation warehouse can run it against yesterday's verdicts without new infrastructure.

I. Against a real stream processor

Everything so far grades reconcilers written for this paper. The same workloads were also replayed through Apache Flink 1.20, as an ordinary LEFT interval join between the two streams, the shape the pattern is usually built as, with the engine's own verdicts graded by the same checker. The decision instant is recovered from `CURRENT_WATERMARK`, which the job selects alongside each row, mapped back to the wall-clock instant that watermark could first have been reached.

The engine violates S1 on 14.91% to 20.20% of obligations, averaged over the eight runs of each window in Table VI; individual runs span 14.78% to 23.01%. It does so in the configuration a competent engineer would write. Widening the join interval buys timeliness and costs soundness, which is the frontier of Section VII-C seen from inside a real system. The engine also emits one output row per matching pair, so a duplicated bank record produces a duplicated match, 15 to 44 extra rows per run, which a downstream sink has to deduplicate on its own.

The window effect is solid; the individual rates are less so. At windows 80 and 320 all eight runs of each window agree to the second decimal place within a seed, across four idleness settings that cannot matter. At window 160 seed 2 is equally stable while seed 1 moves between 18.26% and 20.87% across those same inert settings, which is the engine disagreeing with itself rather than any effect of the configuration. The ordering across windows survives every run we have made; a single per-configuration rate does not. What follows is less comfortable.

Our watermark arms are not a faithful model of the engine. The simulated merged watermark agrees with Flink's partition of the obligations on 77.4% and 78.8% of them, across the two seeds. Roughly a fifth of the time the model and the engine reach different conclusions, and the explanation is structural rather than statistical: the simulated arm models the *decision rule*, raise a break once a watermark has passed, while the engine's interval join also constrains which bank records are eligible to match at all, through the event-time interval in the join predicate. Two different constructions, agreeing most of the time. The timing results in Table III are therefore properties of the models, and we state them as such rather than as measurements of Flink.

The instability rate is itself a random variable. An earlier full run of this experiment, made while the machine was also running the other seven, showed the engine disagreeing with itself: repeating an identical job four times, 26.1% of obligations on one seed and 40.7% on the other received different verdicts. Re-run on an otherwise idle machine, the same configuration gave 0.9% and 0.0%. We withdrew the observation, and the obvious reading of the withdrawal — that load explained it — is one we can now test rather than assert. E10 varies load deliberately, running the determinism check idle and under twelve busy-loop workers on twelve cores, six times per condition.

Load does not explain it. Across six measurements per condition the idle arm spans 0.0% to 21.74% and the loaded arm 0.0% to 25.22%; the condition means differ by 1.16 points, which is an order of magnitude smaller than the spread within either condition. The distributions overlap almost completely, and the largest idle reading exceeds five of the six loaded ones. The useful result is the spread, not the comparison: repeating an identical job on an unchanged machine gives a verdict-instability rate anywhere between zero and a quarter of all obligations, so no single execution of this check — the loaded one we withdrew, the idle one we replaced it with, or any of the twelve reported here — carries information about the engine on its own. The E7 run shipped with this paper is itself one of the high draws: seed 1 disagrees with itself on 21.74% of obligations there, which is why Table VI reports a range per window rather than a figure per configuration.

What that does and does not undermine. The volatility is confined to the repeat-to-repeat determinism check. The violation rates underlying Table III are averages over 200

obligations rather than counts of disagreement between four repeats, and they hold across whole-experiment runs made days apart. The model-fidelity figures sit between the two: reported per seed from one execution, and E10's spread is a reason to read them as approximate. Where this paper quotes a determinism or fidelity number we mean *this run gave this*, not *the engine does this* — so a reviewer who runs our determinism check once and gets a number unlike ours has reproduced us correctly.

One thing E7 could not test at all. Flink's idleness timeout is measured in processing time, and a bounded file source is consumed in milliseconds, so no source is ever idle long enough for the setting to engage. That is why three of its four values give identical numbers above. Proposition 2 is supported by its proof and by the model in Section VII-E, and is *not* confirmed on the engine here. Testing it would need a source paced in processing time, which means a broker and a producer rather than a file, and that is left as future work.

Running the same job with `execution.runtime-mode = BATCH` produces no violations at all, because a batch job sees its whole input and so has no partial-evidence problem to get wrong. Offline reconciliation is sound, and the price is that it has no answer until the batch runs. Everything in this paper is about the systems built to avoid paying that price.

VIII. Discussion

a) A horizon is not a watermark: A watermark summarises what a stream processor has seen. A horizon states what a source promised. They coincide when one source is in play and diverge as soon as two sources have different latencies, because a merged watermark answers "how far has the input progressed" when the question is "could the bank still be about to tell me". Most of the failures in this paper are that substitution.

b) Declaring a bound is the useful part: The typed horizon is not a sophisticated reconciler. It waits for one number and matches on identity. Its advantage is that the number is written down, so when it is wrong the checker says so and names the source. Every other arm encodes the same assumption implicitly in a timeout, where it cannot be checked and cannot be blamed.

c) The check is cheap where the data already is: Nothing in S1 or S2 requires the reconciler's cooperation. Both are predicates over a verdict log, an obligation table and a declared model, which is why Section IV-B can state them as a join. An organisation that cannot change its reconciliation engine, the common case since that engine is often a vendor product, can still run the check against yesterday's output and learn whether its break queue is the engine's doing or its counterparties'.

d) Matching is downstream: A reconciliation product is usually sold on its matching rules. The results here put that second: attribute matching cannot satisfy S2 at any deadline, and identity matching satisfies it immediately. The hard part is not deciding which records correspond. It is deciding when you are allowed to conclude that none does.

IX. Limitations

Two sources, not many. Reconciling three or more generalises the contract, with the horizon becoming a vector, but is not tested, and multi-way reconciliation may raise ambiguities that two-way does not.

Bounds are declared, not learned. Estimating Λ_s online, and stating a confidence rather than a guarantee, is the obvious next step and would turn S1 into a probabilistic property. We chose the bounded form because it is checkable exactly.

Obligations are ledger-side. A bank credit with no ledger expectation is counted and reported but not graded, and that direction of break is a real concern we do not measure.

The workload is synthetic throughout, built from one simulator. Its lateness structure, a fast ledger and a batching bank, is realistic in shape, and the contract does not depend on the particular numbers, but the rates reported here are properties of this generator and should not be read as field rates.

The cost constants are argued, not measured. Where the ordering of arms depends on them, as it does under shift, we report the dependence instead of a figure.

X. Conclusion

Reconciliation on a stream is a decision under partial evidence, and the question of when the evidence is sufficient has been answered in deployed systems by a constant nobody writes down. Writing it down as a per-source lateness bound, with a safety property relating it to the instant a verdict is issued, costs little and buys a diagnosis: contract violation, model violation, or genuine loss, told apart without knowing the answer. The parts of this that surprised us were that the documented remedy for a stalled streaming join is unsound for reconciliation below the slow source's posting period, and that under distribution shift every reconciler we tested was entitled to every verdict it got wrong.

Artifact availability

Code, workload generator, Docker configuration and every results file are at https://github.com/abhisheksharma2411/reconstream. An archival DOI is forthcoming. `make reproduce` runs the whole pipeline from a clean checkout: it fetches FinalityBench, starts PostgreSQL and Flink, runs every experiment, regenerates each figure from the raw results and rebuilds this paper. No number in the text is entered by hand.

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